

On-the-fly Modulation for Balanced Multimodal Learning (OGM & OPM) — Poori Explanation

Calculation, Derivatives, Gradient Descent, Modality Bias Mitigation aur Techniques — Hinglish notes

Paper: Yake Wei, Di Hu, Henghui Du, Ji-Rong Wen — “On-the-fly Modulation for Balanced Multimodal Learning”, IEEE TPAMI 2024 (arXiv:2410.11582). Ye CVPR 2022 wale **OGM-GE** paper (“Balanced Multimodal Learning via On-the-fly Gradient Modulation”) ka extended version hai.

Source file: OGM_DL.pdf

Contents

1. Pehle kya galti ho rahi thi? (The Problem)
2. Model ka Math (Late Fusion) — Eq 1, Eq 2
3. Derivatives & Gradient Descent — Eq 3, 4, 5 + numeric example
4. Discrepancy kaise measure karte hain — Eq 6, 7
5. OPM — On-the-fly Prediction Modulation — Eq 8
6. OGM — On-the-fly Gradient Modulation + GE — Eq 9-17
7. OGM* — weak modality ka gradient badhana kyun kaam nahi karta
8. Results (short)
9. Saari techniques ek jagah
10. Ek line me intuition
11. Cheat-sheet: saare equations ek table me

Notation (symbols ka matlab)

Symbol	Matlab
M, m	Total modalities ki sankhya; m = koi ek modality (e.g., audio / visual)
$x_i = (x_i^1, \dots, x_i^M), y_i$	i -th sample ke M inputs aur uska label; C = classes ki sankhya
$\phi^m(\theta^m, x^m)$	Modality m ka encoder (parameters θ^m), output feature (dimension d_m)
$W = [W^1; \dots; W^M], b$	Final linear classifier ke weights (blocks me tode hue) aur bias
$f(x_i) \in \mathbb{R}^C$	Logits (softmax se pehle ka output)
$\ell(x_i, y_i), L$	Ek sample ka cross-entropy loss; L = poore dataset ka average loss
$\eta, N, B_t, B_t $	Learning rate; total samples; t -th iteration ka mini-batch; batch size
s_i^m, ρ_t^m	

	Modality m ka uni-modal score (correct class ki probability); discrepancy ratio
q_t^m, k_t^m	OPM ka drop probability; OGM ka gradient modulation coefficient
$q_{\text{base}}, \lambda, \alpha$	Hyper-parameters (OPM: q_{base}, λ ; OGM: α)
$z(x) = \tanh(x - 1)$	Monotonic function jo discrepancy ko (0, 1) me map karta hai
$\tilde{g}(\theta), \nabla L(\theta), U$	Mini-batch gradient; full gradient; SGD sampling se aane wali covariance

1. Pehle kya galti ho rahi thi? (The Problem)

Setup: Audio + Visual jaisa multimodal model, dono encoders ko ek saath ek hi loss (cross-entropy) pe train karte ho = **joint training**. Sabse common tareeka: dono features concat karo, ek linear classifier lagao, ek loss.

Observation (Fig 1, VGGSound dataset):

- Audio-visual model > audio-only > visual-only — ye toh expected hai ✓
- **Lekin** jab authors ne audio-visual model ke andar ka **visual encoder** alag se nikaal ke check kiya (encoder freeze karke uske upar naya uni-modal classifier fit kiya), toh wo **visual-only model se worse** nikla. Audio encoder bhi thoda worse tha, par visual **bahut zyada** worse.

Matlab: Joint training me uni-modal encoders **under-optimized** reh jaate hain, aur ye under-optimization **imbalanced** hai — jo modality dataset me zyada “discriminative” hai (VGGSound sound-oriented hai → audio) wo **dominate** karti hai aur doosri (visual) ko **suppress** kar deti hai. Isko hi paper me “imbalanced multimodal learning” / modality bias bolte hain.

Galti kya thi: Ek **uniform objective** sabke liye. Koi mechanism nahi jo dekhe “bhai visual toh seekh hi nahi raha”. Literature me isko alag-alag naam se pakda gaya:

- **Different convergence rates** (Wang et al., CVPR 2020): audio jaldi converge/overfit karta hai, visual dheere; kabhi-kabhi multimodal model best uni-modal se bhi peeche reh jaata hai.
- **Modality Competition** (Huang et al., 2022): theoretically prove kiya ki joint training me sirf ek subset of modalities sufficient features seekh paati hai.
- **Greedy learning** (Wu et al., ICML 2022): model “greedy” hoke jo modality jaldi loss girati hai usi pe depend ho jaata hai.

2. Model ka Math (Late Fusion)

M modalities, har ek ka encoder $\phi^m(\theta^m, x^m)$, features **concat** karo, ek linear classifier (W, b), loss = cross-entropy.

$$f(x_i) = W \cdot [\phi^1(\theta^1, x_i^1); \phi^2(\theta^2, x_i^2); \dots; \phi^M(\theta^M, x_i^M)] + b$$

Eq 1

W ko M column-blocks me todo ($W^m \in \mathbb{R}^{C \times d_m}$) — concat ke saath matrix multiply wahi hai jo alag-alag blocks ka sum hai:

$$f(x_i) = W^1 \cdot \phi_i^1 + W^2 \cdot \phi_i^2 + \dots + W^M \cdot \phi_i^M + b$$

Eq 2 (sabse important)

Yaani **final logits = uni-modal contributions ka SUM**. Audio ka apna “vote” ($W^a \cdot \phi^a$), visual ka apna “vote” ($W^v \cdot \phi^v$), aur bas add ho jaate hain.

Feed-forward stage ki problem

Agar audio ka vote already correct class pe bada hai, toh visual ka vote galat bhi ho toh **sum sahi aa jaata hai**. Audio visual ki galti “correct” kar deta hai → model ko visual sudhaarne ki zaroorat hi mehsoos nahi hoti. Prediction dominant modality decide karti hai.

3. Derivatives & Gradient Descent (Back-propagation analysis)

3.1 Cross-entropy loss

$$\ell(x_i, y_i) = -\log(e^{f_y} / \sum_{c=1..C} e^{f_c}) = -f_y + \log \sum_c e^{f_c}$$

(f_c = class c ka logit, y = true class. $\log(a/b) = \log a - \log b$ use kiya.)

3.2 Logits ke respect me derivative (Eq 5 ka derivation)

- **Term 1:** $\partial(-f_y)/\partial f_c = -1$ agar $c = y$, warna $0 \rightarrow -1[c = y]$
- **Term 2:** $\partial/\partial f_c [\log \sum_j e^{f_j}] = (1 / \sum_j e^{f_j}) \cdot e^{f_c} = \text{softmax}(f)_c$

$$\partial \ell / \partial f_c = \text{softmax}(f)_c - 1[c = y] \quad (\text{vector form: } \partial \ell / \partial f = p - \text{onehot}(y), p = \text{softmax}(f))$$

Eq 5

Ye famous “softmax minus one-hot” result hai. Paper me softmax ka numerator explicitly likha hai: $e^{(W^1 \cdot \phi^1 + \dots + W^M \cdot \phi^M + b)_c}$ — yaani numerator aur denominator dono **sabki sum** se bante hain.

3.3 Chain rule se modality m ke parameters tak

Kyunki $f = \sum_m W^m \phi^m + b$, isliye $\partial f / \partial W^m$ me sirf ϕ^m aata hai aur $\partial f / \partial \theta^m = W^m \cdot \partial \phi^m / \partial \theta^m$.

Gradient Descent (GD) — poore N samples ka average — se update:

$$W_{t+1}^m = W_t^m - \eta \cdot (1/N) \sum_{i=1..N} \nabla_{W^m} \ell(x_i, y_i) = W_t^m - \eta \cdot (1/N) \sum_i (\partial \ell_i / \partial f) \cdot \phi_{i,t}^m$$

Eq 3 (classifier block)

Eq 4 (encoder)

$$\theta_{t+1}^m = \theta_t^m - \eta \cdot (1/N) \sum_{i=1..N} \nabla_{\theta^m} \ell(x_i, y_i) = \theta_t^m - \eta \cdot (1/N) \sum_i (\partial \ell_i / \partial \mathbf{f}) \cdot \partial (W_t^m \cdot \phi_{i,t}^m) / \partial \theta_t^m$$

($\nabla_{W^m} \ell$ actually outer product hai: $(\mathbf{p} - \text{onehot}(y)) (\phi^m)^T$, size $C \times d_m$. Iska magnitude $\|\mathbf{p} - \text{onehot}(y)\|$ ke proportional hai.)

Key insight: Dono equations me modality m ka update **sirf ek term** ke through doosri modalities se juda hai: $\partial \ell / \partial \mathbf{f} = \text{softmax}(\mathbf{f}) - \text{onehot}(\mathbf{y})$. Baaki sab $(\phi^m, W^m, \partial \phi^m / \partial \theta^m)$ modality m ka apna hai. Aur softmax($\mathbf{f}$) **sabke sum** se banta hai.

Toh agar dominant modality ne correct class ka logit bahut bada kar diya $\rightarrow \text{softmax}(\mathbf{f})_y \approx 1 \rightarrow \partial \ell / \partial \mathbf{f} \approx \mathbf{0} \rightarrow$ **sabka gradient chota**, weak modality ka bhi. Weak modality ko “optimization effort” milta hi nahi, jabki uska apna feature abhi bhi kharab hai. **Yahi under-optimization hai.** Encoders ek doosre se feedback leke adjust nahi kar sakte — sirf shared loss term hai.

3.4 Chota numeric example

3 classes, true class = 0. Audio strong: $W^a \cdot \phi^a = [4, 0, 0]$. Visual weak: $W^v \cdot \phi^v = [0.5, 0.3, 0.2]$ (bias = 0 maan lo).

	logits $\mathbf{f}$	softmax $\mathbf{p}$	$\partial \ell / \partial \mathbf{f} = \mathbf{p} - \text{onehot}$	loss
Joint (audio + visual)	[4.5, 0.3, 0.2]	[0.972, 0.015, 0.013]	[-0.028, 0.015, 0.013]	0.028
Visual-only	[0.5, 0.3, 0.2]	[0.391, 0.320, 0.289]	[-0.609, 0.320, 0.289]	0.94

Visual encoder ko joint training me **~22x chota gradient** milta hai ($\|\partial \ell / \partial \mathbf{f}\|$ ka ratio), sirf isliye ki audio ne loss pehle hi gira diya. Visual “correct” hai (argmax sahi), par actually usne kuch khaas seekha nahi — aur aage bhi nahi seekhega kyunki signal hi nahi aa raha.

3.5 GD vs SGD (baad me OGM me kaam aayega)

- **GD (Eq 9):** $\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t)$, jahan $\nabla L = (1/N) \sum_i \nabla \ell(x_i)$ = poore dataset ka gradient. Har step me saara data — practically bahut mehenga.
- **SGD (Eq 10):** $\theta_{t+1} = \theta_t - \eta \tilde{g}(\theta_t)$, jahan $\tilde{g} = (1/|B_t|) \sum_{x \in B_t} \nabla \ell(x)$ = sirf mini-batch ka gradient. Ye full gradient ka **unbiased estimate** hai (expected value same), par har batch me thoda random error hota hai — isi ko “SGD noise” bolte hain.

4. Discrepancy kaise measure karte hain (dono methods ka common part)

Modulation karne se pehle pata hona chahiye **kaunsi modality dominant hai aur kitni**. Iske liye training ke dauran, har mini-batch pe, bina koi extra network ke:

Eq 6 — uni-modal score

$$s_i^m = \sum_c 1[c = y_i] \cdot \text{softmax}(W^m \cdot \phi_i^m + b/M)_c = \text{softmax}(W^m \cdot \phi_i^m + b/M)_{y_i}$$

→ Modality m **akele** correct class ko kitni probability deta hai. Eq 2 ka apna block hi uska “approximate uni-modal prediction” hai; bias ko M parts me equally split kiya (ablation Table 14: split kaise bhi karo, fark nahi padta kyunki bias chhota hota hai).

Eq 7 — discrepancy ratio

$$\rho_t^m = (1/(M-1)) \cdot \sum_{j \neq m} [\sum_{i \in B_t} s_i^m / \sum_{i \in B_t} s_i^j]$$

- Mini-batch pe modality m ka total score ÷ doosri modality ka total score, phir sab doosri modalities pe average.
- $\rho^m > 1 \rightarrow$ modality m **dominant** hai; $\rho^m \leq 1 \rightarrow$ modality m **weak** hai.
- 2 modalities ke case me $\rho^{\text{audio}} = 1 / \rho^{\text{visual}}$.

Upar wale example me: $s_{\text{audio}} = 0.965$, $s_{\text{visual}} = 0.391 \rightarrow \rho_{\text{audio}} = 2.47$, $\rho_{\text{visual}} = 0.41$.

$z(x) = \tanh(x - 1)$: ρ ko (0, 1) me map karta hai. $\rho \approx 1 \rightarrow z \approx 0$ (koi modulation nahi); ρ bada $\rightarrow z \rightarrow 1$ (strong modulation). Example: $\tanh(2.47 - 1) = 0.899$. (Ablation Table 15: sigmoid(x) bhi chalta hai, ratio ki jagah difference bhi chalta hai — idea robust hai.)

5. OPM — On-the-fly Prediction Modulation (feed-forward stage fix)

Idea: Dominant modality ka feature **randomly drop** kar do (zero kar do) with probability q . Jab audio drop hota hai, visual ko **akele** predict karna padta hai $\rightarrow$ loss bada $\rightarrow \partial \ell / \partial f$ bada $\rightarrow$ **visual ko poora gradient** milta hai (upar wali table ka “Visual-only” row wala case).

Eq 8 — adaptive drop probability

$$q_{t+1}^m = q_{\text{base}} \cdot (1 + \lambda \cdot z(\rho_t^m)) \quad \text{agar } \rho_t^m > 1; \quad \text{warna } 0$$

- Range: $(q_{\text{base}}, q_{\text{base}} \cdot (1 + \lambda))$; practically max 1 pe clamp.
- $q_{\text{base}} \in (0, 1)$ drop probability ka range set karta hai; $\lambda > 0$ adjustment ki degree.
- **Weak modality kabhi drop nahi hoti** ($q = 0$) — use har baar training milti hai.
- Discrepancy jitni badi, drop probability utni zyada. $\rho \approx 1 \rightarrow q \approx q_{\text{base}}$.
- Example: $q_{\text{base}} = 0.5$, $\lambda = 0.5$, $\rho = 2.47 \rightarrow z = 0.899 \rightarrow q_{\text{audio}} \approx 0.73$, $q_{\text{visual}} = 0$.

Algorithm 1 (OPM): har iteration t : mini-batch B_t lo $\rightarrow$ forward pass $\rightarrow$ Eq 6, 7 se ρ_t^m nikalo $\rightarrow$ Eq 8 se q_{t+1}^m update karo $\rightarrow$ har modality ka feature probability q_t^m se drop karo $\rightarrow$ backprop $\rightarrow$ parameters update.

ModDrop se farak: ModDrop (Neverova et al.) me har modality ki **fixed** drop probability hoti hai poori training me; OPM me har iteration discrepancy dekh ke **adaptive** hoti hai (Fig 7: audio ka drop prob 0.66–0.70 ke beech dynamically badalta raha, visual ka 0 raha). Isliye OPM $>$ ModDrop (Table 6).

Bonus: Dropout gradient me extra noise laata hai → wider minima → better generalization. Aur kyunki training me model ne “ek modality missing” wala case dekha hai, test time pe **missing modality** ke against bhi robust hai (Fig 6).

6. OGM — On-the-fly Gradient Modulation (back-prop stage fix)

6.1 Gradient ko modulate karna

Eq 9 (GD)

$$\theta^m_{t+1} = \theta^m_t - \eta \nabla_{\theta^m} L(\theta^m_t), \quad \nabla L = (1/N) \sum_i \nabla \ell(x_i; \theta^m_t) \text{ (full gradient)}$$

Eq 10 (SGD)

$$\theta^m_{t+1} = \theta^m_t - \eta \tilde{g}(\theta^m_t), \quad \tilde{g} = (1/|B_t|) \sum_{x \in B_t} \nabla \ell(x; \theta^m_t) \text{ (mini-batch gradient, unbiased estimate)}$$

Eq 11 — modulation coefficient

$$k^m_t = 1 - \alpha \cdot z(\rho^m_t) \text{ agar } \rho^m_t > 1; \text{ warna } \mathbf{1}$$

- Range: $(1-\alpha, 1)$; practically min 0 pe clamp. $\alpha > 0$ modulation ki degree.
- Weak modality ka $k = 1 \rightarrow$ uska gradient **bilkul touch nahi** hota.
- Example: $\alpha = 0.5, \rho = 2.47 \rightarrow k_{\text{audio}} = \mathbf{0.55}, k_{\text{visual}} = 1$.

Eq 12 — modulated update

$$\theta^m_{t+1} = \theta^m_t - \eta \cdot \mathbf{k}^m_t \cdot \tilde{g}(\theta^m_t)$$

Dominant modality ka gradient **scale down** → wo dheere seekhti hai → loss dheere girta hai → $\partial \ell / \partial \theta$ zyada der tak bada rehta hai → weak modality ko zyada aur behtar training milti hai. Overall training thodi slow hoti hai (Fig 4), par final result better.

6.2 GE part — gradient scale down karne se generalization kyun hurt hoti hai, aur fix

Eq 13 (CLT)

$$\tilde{g}(\theta^m_t) \sim N(\nabla_{\theta^m} L(\theta^m_t), U / |B_t|)$$

Mini-batch gradient = full gradient + random error. Bade batch ke liye Central Limit Theorem se ye error approx Gaussian hai; U = sampling covariance. Batch jitna bada, variance utni kam.

Eq 14 — SGD noise

$$\theta^m_{t+1} = \theta^m_t - \eta \nabla L(\theta^m_t) + \xi_t, \quad \xi_t \sim N(0, \eta^2 U / |B_t|)$$

→ Ye **SGD noise** hai; strength $\propto \eta / |B_t|$. Literature (Jastrzebski et al., Zhu et al.) kehta hai zyada noise = wider/flatter minima = **better generalization**, aur saddle points se nikalne me madad. Paper ne **Table 1** me verify kiya:

Setting	CREMA-D Acc	Kinetics-Sounds Acc
$ B = 32, \eta = 1e-5$	40.5	47.6
$ B = 32, \eta = 1e-4$	63.4	61.9
$ B = 32, \eta = 1e-3$ (zyada noise)	66.9	63.0
$ B = 64, \eta = 1e-3$	64.7	62.1
$ B = 128, \eta = 1e-3$ (kam noise)	61.9	58.7

Eq 15 — problem

$$\theta_{t+1}^m = \theta_t^m - \eta k_t^m \nabla L + \xi'_t, \quad \xi'_t \sim N(0, (k_t^m)^2 \cdot \eta^2 U / |B_t|)$$

$k < 1 \rightarrow$ noise variance k^2 se kam ho gayi ($k = 0.5$ pe sirf 25% bachi) $\rightarrow$ generalization risk. Sirf gradient scale karna kaafi nahi.

Eq 16 — fix (Generalization Enhancement)

$$\theta_{t+1}^m = \theta_t^m - \eta \cdot (k_t^m \tilde{g}(\theta_t^m) + h(\theta_t^m)), \quad h \sim N(0, U / |B_t|) \rightarrow \text{extra term } \varepsilon_t \sim N(0, \eta^2 U / |B_t|)$$

Zero-mean Gaussian noise add karo jiska covariance current mini-batch gradient jaisa hi hai. ε_t aur ξ'_t independent hain, toh variances add ho jaati hain:

Eq 17 — final

$$\theta_{t+1}^m = \theta_t^m - \eta k_t^m \nabla L(\theta_t^m) + \xi''_t, \quad \xi''_t \sim N(0, ((k_t^m)^2 + 1) \cdot \eta^2 U / |B_t|)$$

$k^2 + 1 > 1$ hamesha $\rightarrow$ noise sirf **recover nahi, enhance** ho gayi original SGD se bhi zyada. Isliye original paper ka naam “OGM-GE” tha.

k	noise variance factor bina GE (k^2)	GE ke saath ($k^2 + 1$)
1.0 (koi modulation nahi)	1.00	2.00
0.8	0.64	1.64
0.5	0.25	1.25

Algorithm 2 (OGM): har iteration t : mini-batch B_t lo $\rightarrow$ forward $\rightarrow$ Eq 6, 7 se $p_t^m \rightarrow$ Eq 11 se $k_t^m \rightarrow$ backprop se $\tilde{g}(\theta_t^m) \rightarrow \tilde{g}$ ki covariance ke basis pe $h(\theta_t^m)$ sample karo $\rightarrow \theta_{t+1}^m = \theta_t^m - \eta (k_t^m \tilde{g} + h)$.

Practical note: Official implementation me h ka std gradient tensor ke std ke barabar rakha jaata hai ($\text{grad} * k + N(0, \text{grad.std}())$), modulation encoder ke parameters ke gradients pe lagta hai, aur k / q har mini-batch pe recompute hote hain — koi extra network ya loss nahi, isliye training time sirf $\sim 1.01\times$.

7. OGM* — “Weak modality ka gradient badha kyun nahi dete?”

Authors ne ye ulta idea bhi try kiya — **OGM***: dominant modality ka gradient waise hi rehne do, aur **weak** modality ka gradient discrepancy ke hisaab se **badha** do.

Method	CREMA-D Acc	CREMA-D mAP	Kinetics-Sounds Acc	Kinetics-Sounds mAP
Concatenation (baseline)	66.9	72.1	63.0	67.3
OGM (dominant ko slow karo)	74.6	81.4	65.4	71.4
OGM* (weak ko boost karo)	67.2	72.7	63.9	67.8

Kyun fail hota hai: Dominant modality ab bhi easy-to-learn hai aur dominate karti rehti hai — discrepancy ratio high hi rehta hai (Fig 5b; CentralNet jaise complex model me bhi kam nahi hota). Plus boosted gradient global view me direction/scale dono me galat ho sakta hai.

Dominant ko slow karna kaam karta hai, weak ko push karna nahi.

8. Results (short)

Method (Acc %)	CREMA-D (Audio+Visual)	Kinetics-Sounds (Audio+Visual)	UCF-101 (RGB+Optical Flow)	VGGSound (Audio+Visual)
Audio / RGB-only (individually trained)	61.4	49.9	77.2	47.5
Visual / OF-only (individually trained)	50.9	46.7	59.9	28.7
Concatenation (joint, baseline)	66.9	63.0	80.5	52.7
Concatenation + OPM	75.1	67.0	81.9	54.1
Concatenation + OGM	74.6	65.4	81.5	53.6
Concatenation + OPM + OGM	76.7	68.0	81.5	53.3

- **Failure case dikhta hai:** CREMA-D pe Summation fusion (60.8) audio-only (61.4) se bhi peeche — joint training kabhi-kabhi best uni-modal se bhi worse.
- **OPM > OGM** usually — kyunki dropping zyada strong modulation hai (weak modality temporarily poori prediction control karti hai).
- **OPM + OGM** kabhi better, kabhi nahi — dropping se discrepancy estimate change ho jaata hai, toh OGM ka k thoda off ho jaata hai. Authors ne future work bola.
- **Training time** sirf 1.01× (Table 4); G-Blending 1.50× kyunki extra uni-modal classifiers train karta hai.

- **Versatility:** Summation, FiLM, CentralNet, VATT, MMTM (cross-modal interaction), TSN, TSM, I-vector/X-vector (task-oriented), multi-layer classifier (yahan s^m “zero-out” strategy se estimate: doosri modality ka feature 0 karke prediction dekho), 3 modalities (CMU-MOSI, UCF-101-Three), AVE event localization, MUSIC-AVQA — sab jagah gain.
- **Loss curve (Fig 4):** OPM/OGM ka final training loss baseline se higher — matlab training samples ko over-memorize kam kar raha hai.
- **Discrepancy ratio (Fig 5):** modulation ke baad clearly girta hai, par 1 tak nahi pahunchta — dataset me modalities ki information naturally barabar nahi hoti.
- **Category-wise (Fig 8):** jis category me jo modality weak thi, usi ko zyada gain; chahe audio dominant ho (playing saxophone) ya visual (dribbling basketball).
- **Ablations:** tanh vs sigmoid, ratio vs difference, bias split (γ), λ / q_{base} / α (0.2–0.6) — sab robust, har choice baseline se better.

9. Saari techniques ek jagah (related work + is paper ki)

Technique	Kaise kaam karti hai	Limitation / note
G-Blending (Wang et al., CVPR 2020)	Multimodal loss ke saath extra uni-modal losses; weights “overfitting-to-generalization ratio” se decide	Extra uni-modal classifiers chahiye, $\sim 1.5\times$ training time
Greedy / Conditional utilization rate (Wu et al., ICML 2022)	Measure karo model kis modality se kitni speed se seekh raha hai; underutilized modality ko extra training steps	Extra training steps
Uni-modal teachers / KD (Du et al., 2021)	Pehle se trained uni-modal encoders se multimodal model me knowledge distillation	Pehle alag uni-modal models train karne padte hain
Adaptive tracking factor (Sun et al., 2021)	Har modality ka learning rate adaptively balance	—
ModDrop (Neverova et al., 2015)	Fixed probability se poori modality drop (originally robustness ke liye)	Fixed prob dynamic learning pace ko match nahi karta; phir bhi baseline se better
Neuron dropout variants (Annealed / Evolutional / Curriculum dropout)	Neuron-level dropout schedules	Modality-level discrepancy address nahi karte — Table 6 me koi gain nahi
Shapley sample-level valuation (Wei et al., CVPR 2024)	Har sample pe har modality ka contribution Shapley value se measure karke fine-grained balance	—
Diagnosing & Re-learning (Wei et al., ECCV 2024)	Modality ki limited capacity consider karke encoder re-initialization se uni-modal learning control	—
MMPareto (Wei & Hu, ICML 2024)	Uni-modal aur multimodal gradients ko Pareto-style integrate karna	—
Robustness view (Yang et al., 2024)	Imbalance ka robustness pe effect, enhancement strategy	—
OPM (ye paper)	Dominant modality ka feature adaptive probability se drop (feed-forward stage)	OGM ke saath combine karne pe discrepancy estimate disturb
OGM + GE (ye paper / CVPR 2022)	Dominant modality ka gradient k se scale down + Gaussian noise se generalization recover (back-propagation stage)	OPM se thoda weak modulation

10. Ek line me intuition

Group project socho: **topper (audio)** sab kaam kar leta hai, marks (loss) achhe aate hain, toh **weak student (visual)** ko kabhi seekhna hi nahi padta — teacher (gradient) ko lagta hai sab theek hai.

- **OPM** = topper ko kabhi-kabhi chhutti pe bhej do → weak student ko khud karna padega, aur uski galtiyon pe direct feedback milega.
- **OGM** = topper ki speed slow kar do ($\text{gradient} \times k$) taaki weak student catch up kare.
- **GE** = speed slow karne se exploration (SGD noise) kam hui, toh thoda randomness (Gaussian noise) wapas daal do.
- **OGM*** (weak ko push karna) kaam nahi karta kyunki topper phir bhi sab kar leta hai.

11. Cheat-sheet: saare equations ek table me

Eq	Naam	Formula	Role
1	Logits (concat)	$f(x) = W[\phi^1; \dots; \phi^M] + b$	Late-fusion model
2	Logits (sum form)	$f(x) = \sum_m W^m \cdot \phi^m + b$	Prediction = uni-modal votes ka sum
3	W^m update (GD)	$W^m \leftarrow W^m - \eta (1/N) \sum_i (\partial \ell_i / \partial f) \phi_i^m$	Cross-modal link sirf $\partial \ell / \partial f$ se
4	θ^m update (GD)	$\theta^m \leftarrow \theta^m - \eta (1/N) \sum_i (\partial \ell_i / \partial f) \partial (W^m \phi_i^m) / \partial \theta^m$	Encoder ka gradient
5	CE derivative	$\partial \ell / \partial f_c = \text{softmax}(f)_c - 1[c = y]$	Dominant modality ise ≈ 0 kar deti hai
6	Uni-modal score	$s_i^m = \text{softmax}(W^m \phi_i^m + b/M)_{y_i}$	Modality m akele kitna confident
7	Discrepancy ratio	$\rho_t^m = (1/(M-1)) \sum_{j \neq m} s_i^m / s_i^j$	>1 dominant, ≤ 1 weak
8	OPM drop prob	$q^m = q_{\text{base}}(1 + \lambda \tanh(\rho^m - 1))$ if $\rho^m > 1$ else 0	Feed-forward modulation
9 / 10	GD / SGD	$\theta \leftarrow \theta - \eta \nabla L$ / $\theta \leftarrow \theta - \eta \tilde{g}$	Full vs mini-batch gradient
11	OGM coefficient	$k^m = 1 - \alpha \tanh(\rho^m - 1)$ if $\rho^m > 1$ else 1	Back-prop modulation
12	OGM update	$\theta^m \leftarrow \theta^m - \eta k^m \tilde{g}$	Dominant slow
13	CLT	$\tilde{g} \sim N(\nabla L, U/ B)$	Mini-batch gradient Gaussian
14	SGD noise	$\xi \sim N(0, \eta^2 U/ B)$	Noise $\propto \eta/ B \rightarrow$ generalization
15	Modulated noise	$\xi' \sim N(0, k^2 \eta^2 U/ B)$	Problem: noise kam
16	GE noise add	$\theta \leftarrow \theta - \eta(k \tilde{g} + h)$, $h \sim N(0, U/ B)$	Fix
17	Final noise	$\xi'' \sim N(0, (k^2 + 1)\eta^2 U/ B)$	Recovered + enhanced

— End of notes — Based on OGM_DL.pdf (Wei et al., TPAMI 2024). Numeric example values khud calculate kiye gaye hain (3-class toy setup). —